MIT 18.06 SC - [redacted] 1.7

Problem 6.1

a) Let s and t be any two vectors in $S+T$. By definition of $S+T$, ~~$s \in S$ and~~ there exists vectors $s_1 \in S$ and $t_1 \in T$ such that $s_1 + t_1 = s$ and there exists vectors s_2 and t_2 such that $s_2 + t_2 = t'$ (where $s_2 \in S$ and $t_2 \in T$).

Then $s+t = (s_1+t_1)+(s_2+t_2)$

$= (s_1+s_2)+(t_1+t_2)$

Since $s_1, s_2 \in S$ ~~&~~ where S is a ~~subspace of a~~ vector space, ~~&~~ $s_1+s_2 \in S$, and since $t_1, t_2 \in T$ where T is a vector space, $t_1+t_2 \in T$, and so $(s_1+s_2)+(t_1+t_2) = s+t \in S+T$, so $S+T$ is closed under addition. ~~Also, if st is any vector in~~ Also, let x be any scalar, then $x(s_1+s_2) \in S$ and $x(t_1+t_2) \in T$ (since S and T are vector spaces) and so $x(s+t) \in S+T$ so $S+T$ is closed under scalar multiplication. Hence $S+T$ is a vector space

b) $S+T$ is the sum of all vectors $s \in S$ and $t \in T$ (and hence a plane if S and T are distinct). On the other hand, $S \cup T$ is just the two lines S and T. The span of S and T, and hence the span of $S \cup T$, is the set of all linear combinations of vectors in S and T, which, by part (a), forms $S+T$, and so $S+T$ is the span of $S \cup T$

Problem 6.2

$$x - 3y - z = 12 \Rightarrow x = 12 + 3y + z$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 + 3y + z \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Problem 6.3

$$N(C) = N(A) \cap N(B)$$